PARTS LIST

Item	Qty	REFERENCE DESIGNATOR	PART NUMBER	Description	MANUFACTURER
1	1		S1101EFF0313	BARE BOARD	ZIN TECHNOLOGIES
2	11	C3, C6, C7, C8, C9, C10, C11, C12, C13, C14, C15	TR3E106K063C0600	CAPACITOR TANTALUM 10uF 63V 10% SMD 7343	KEMET
3	1	C16	597D476X9050Z2T	CAPACITOR, TANTALUM, 47uF, 50V, 10%, SMD 7360	VISHAY
4	1	C17	C0805C109D1GACTU	CAPACITOR, CERAMIC, 1pF, 100V, NPO, 0805	TDK
5	5	C18, C19, C20, C21, C43	C0805C104K1RACTU	CAPACITOR 0.1uF 100V CERAMIC 10% 0805 SMD	KEMET
6	1	C42	T494B106K016AT	CAPACITOR 10.0uF 16V TANTALUM 10% B-SIZE SMD	KEMET
7	19	D1, D3, D4, D5, D6, D7, D8, D9, D10, D11, D12, D13, D14, D15, D16, D17, D18, D19, D20	CD1206-B260	DIODE SMD SCHOTTKY RECTIFIER 60V 2A 1206	BOURNS
8	1	J1	HTSS-120-01-L-D	CONNECTOR, HDR, SHROUDED, .1 PITCH, .025" SQ POST	SAMTEC
9	9	L1, L2, L3, L4, L5, L6, L7, L8, L9	8532R-37L	INDUCTOR 1000uH POWER SMD	API DELEVAN
10	29	R1, R2, R3, R4, R5, R6, R7, R8, R9, R11, R12, R13, R14, R15, R16, R17, R19, R20, R21, R22, R23, R25, R27, R28, R93, R97, R98, R99, R100	ERJ-8ENF4750V	RESISTOR 475 OHM 1/4W 1% 1206 SMD	PANASONIC - ECG
11	9	R31, R34, R37, R40, R43, R46, R49, R52, R55	RNCP0805FTD10K0	RESISTOR 10.0K 1/4W .1% 0805 SMD	STACKPOLE
12	9	R32, R35, R38, R41, R44, R47, R50, R53, R56	ERJ-6ENF6652V	RESISTOR, 66.5K, 1/8W, 1%, 0805 SMD	PANASONIC - ECG
13	2	R91, R92	ERJ-6ENF1001V	RESISTOR 1.00K 1/8W 1% 0805 SMD	PANASONIC - ECG
14	1	R94	ERJ-6ENF3010V	RESISTOR, 301 OHM, 1/8W, 1%, 0805 SMD	PANASONIC - ECG
15	1	R130	RC0805FR-071M33L	RESISTOR, 1.33M, 1/8W, 1%, 0805 SMD	YAGEO
16	2	U1, U2	LT3080EQ#PBF	ADJUSTABLE VOLTAGE REGULATOR	LINEAR TECHNOLOGIES
17	14	U3, U4, U5, U6, U7, U8, U10, U11, U12, U13, U14, U15, U16, U17	SZD20CF	RELAY, SMT, 2A, 60VDC, OPTO-ISOLATED SHORT-CIRCUIT PROTECTED	TELEDYNE
18	9	U28, U29, U30, U31, U32, U33, U34, U35, U36	DRV102FKTWT	PWM SOLENOID/VALVE DRIVER	TI
19	2	U37, U38	CD74HC245M	IC BUS TRANSCEIVER 8BIT 20SOIC	TEXAS INSTRUMENTS
20	1	U39	CD74HC32M	IC, QUAD 2-INPUT OR GATE, 14 PIN SOIC	TEXAS INSTRUMENTS
21	1	U40	CD74HCT04M	IC, SMT, HEX INVERTERS, 14-SOIC	TEXAS INSTRUMENTS
22	A/R		TZ-211	LABEL, 1/4 BLK LETTERS WHT BACKROUND	BROTHER
23	A/R		3140	RTV FOR CONFORMAL COATING	DOW CORNING
24	A/R		PR-1204	PRIMER FOR CONFORMAL COATING	DOW CORNING
25	A/R		SN63PB37/RMA	SOLDER	
26	9	C44, C45, C46, C47, C48, C49, C50, C51, C52	CGA4J3X7S2A474M125AE	CAP, CER, 0.47uF, 100V, 20%, X7S, 0805	TDK

8. CONFORMAL COATING, STAKING AND INSPECTION SHALL BE IN ACCORDANCE WITH NASA-STD-8739.1 AND THE FOLLOWING NOTES:
   - A. PERFORM CLEANLINESS TEST.
   - B. BAKE PWA AT 93°C ±5.5°C (200°F ±10°F) FOR 4 HOURS PRIOR TO CONFORMAL COATING.
   - △C MASK OFF ALL DESIGNATED AREAS USING KAPTON TAPE. ALL MASKING OPERATIONS SHALL BE PERFORMED UNDER AN AIR IONIZER.
   - △D MEASURE CONFORMAL COATING THICKNESS AT DESIGNATED AREAS.
   - E. CONFORMAL COAT THE COMPLETE ASSEMBLY USING ITEM 23 AND ITEM 24.
9. ENGINEER OR TECHNICIAN SHALL FUNCTIONALLY TEST THE COMPLETED ASSEMBLY AFTER CONFORMAL COATING PER TEST PLAN ACME-TST-028.
10. PLACE THE COMPLETED ASSEMBLY IN CLEAN, STATIC DISSIPATIVE PACKAGING. IDENTIFY PART BY TAGGING WITH PART NUMBER AND SERIAL/LOT NUMBER AS SPECIFIED IN WORK ORDER PER SAE AS478, METHOD 35. STORE IN A PROJECT APPROVED STORAGE AREA.
11. REFERENCE SCHEMATIC: S1101EFS0313
12. EXPEDITION CONTROL FILES: S1101EFA0313_REV_C.ZIP

NOTES: UNLESS OTHERWISE SPECIFIED

1. CERTIFICATE OF CONFORMANCE REQUIRED FOR ALL COMMERCIAL PARTS.
2. ESD HANDLING AND CONTROL PER ESTABLISHED PROCEDURES.
3. HANDLE ALL PARTS AND ASSEMBLIES WITH CLEAN, LINT-FREE, ANTI-STATIC GLOVES OR FINGER COTS.
4. DEMOISTURIZE (BAKE OUT) PWB (ITEM 1) FOR A MINIMUM OF 2 HOURS AT 93°C ±5.5°C (200°F ±10°F.) PERFORM SOLDERING IMMEDIATELY AFTER BAKING, OR STORE IN A CLEAN, DRY, SEALED STATIC DISSIPATIVE BAG OR CONTAINER.
5. △5 MARK THE ASSEMBLY WITH REV AND S/N USING ITEM 22.
6. ASSEMBLY, SOLDERING AND INSPECTION SHALL BE IN ACCORDANCE WITH NASA-STD-8739.2, NASA-STD-8739.3, AND THE FOLLOWING NOTES:
   - A. INSTALL AND SOLDER ALL SURFACE MOUNT (SMT) PART BEFORE THROUGH HOLE PARTS USING SOLDER (ITEM 25.)
   - B. FOR THROUGH HOLE PARTS, LEAD TERMINATIONS SHALL BE STRAIGHT THROUGH (NOT LAPPED OR CLINCHED.)
7. ENGINEER OR TECHNICIAN SHALL FUNCTIONALLY TEST THE COMPLETED ASSEMBLY PRIOR TO CONFORMAL COATING PER TEST PLAN ACME-TST-022.

ACME-ECO-485 In Process

REVISION HISTORY

REV	DESCRIPTION	APPROVED
C	REVISE PER ACME-ECO-424	MH 3/18/15 TG 3/18/15 DS 3/19/15
B	REVISE PER ACME-ECO-317 AND ACME-ECO-325	TAG 9/18/14 DAS 9/16/14 RJP 9/16/14
A	REVISE PER ACME-ECO-274	TAG 4/10/14 DAS 4/10/14 RJP 4/10/14

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S1101MFA0413

NEXT ASSEMBLY

UNLESS OTHERWISE SPECIFIED:
1. DIMENSIONS ARE IN INCHES
2. BREAK EDGES .010-.020
3. INSIDE CORNERS R.02 MAX
4. LINEAR TOLERANCES:
   .X ± 0.03
   .XX ± 0.01
   .XXX ± 0.005
5. ANGULAR TOLERANCES: ± 0.5°

THIRD ANGLE PROJECTION

THIS DRAWING IS UNDER CONFIGURATION CONTROL: DO NOT REVISE WITHOUT AN APPROVED ECO.

APPROVED: E. McCLOSKEY 3/19/15

PRIOR TO USE, CHECK WITH CONFIGURATION MANAGEMENT FOR ANY OUTSTANDING ECOS.

DESIGNER	R. PFEIFFER 7/12/13	QUALITY	D. CHOVAN 7/12/13
REVIEWER	D. STRANG 7/12/13	MATERIALS	M. WILKINSON 7/12/13
MECHANICAL	A. DRAKE 7/16/13	PROJ LEAD	B. BOROWSKI 7/16/13
ELECTRICAL	T. GOBELI 7/12/13	-	-
STRUCTURAL	W. BORRELLI 7/15/13	-	-
SAFETY	C. WERNER 7/12/13	-	-

DIMENSIONING AND TOLERANCING PER ASME Y14.5M-1994

CONTRACT NUMBER: NNC09BA02B
DELIVERY CODE: SPACE110

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CLEVELAND, OHIO

TITLE

ACME
UPSTREAM RELAY BOARD
ASSEMBLY DRAWING

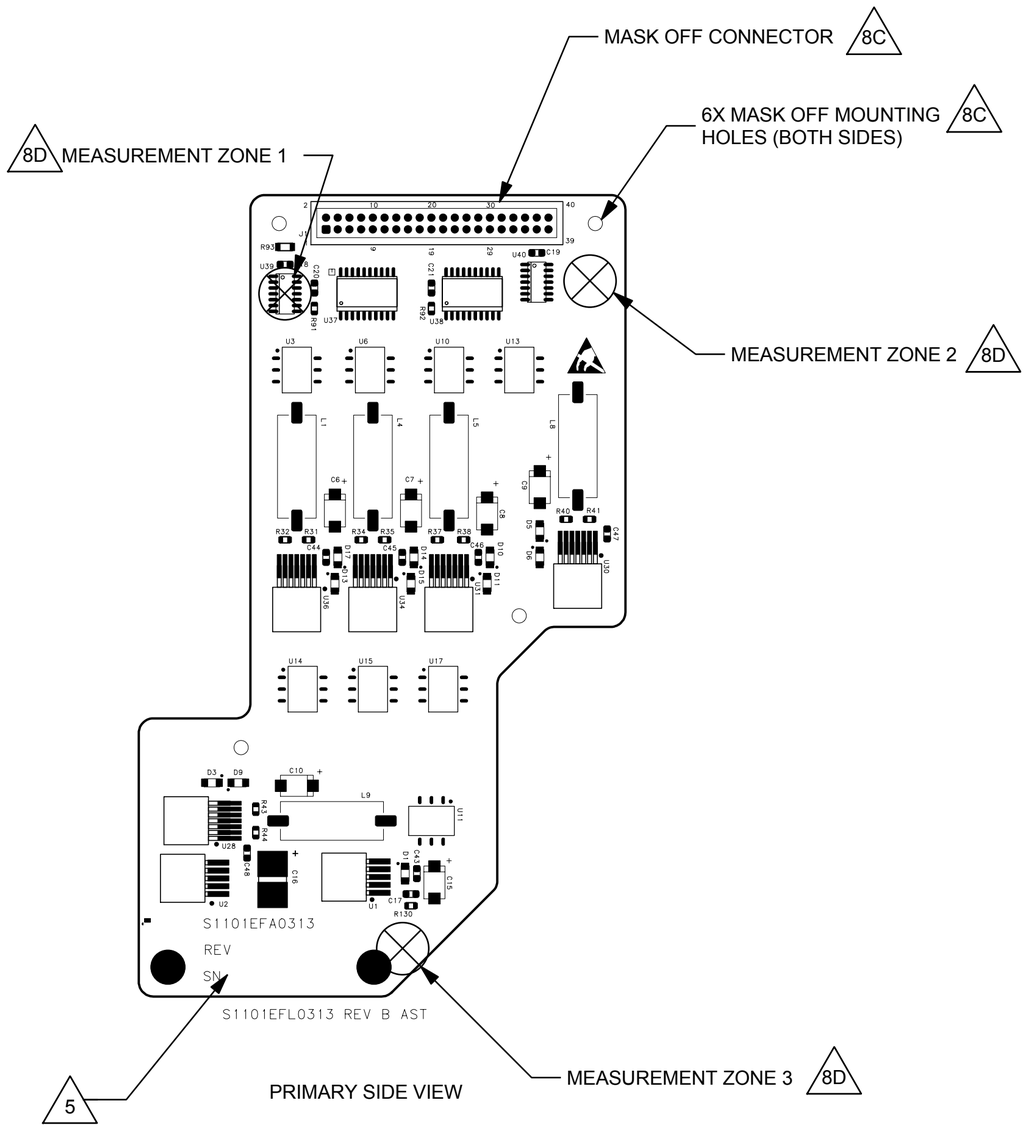
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C	0SG68	S1101EFA0313	C

SCALE: NONE | INITIAL REL DATE 7/16/13 | SHEET 1 OF 2

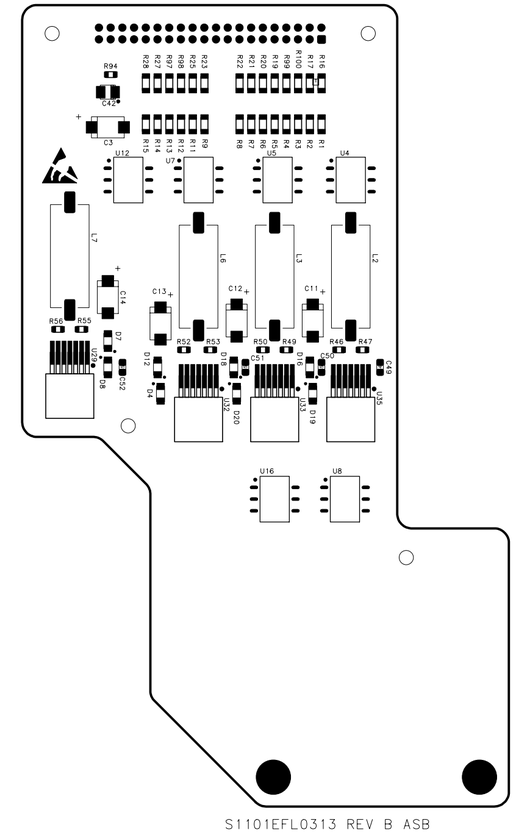
RELEASE LEVEL: .
MODEL NAME: S1101EFA0313_REV_C.bpd

BLUEPRINT
VERSION 3.0

REV C
SH 1
DWG NO S1101EFA0313



PRIMARY SIDE VIEW



SECONDARY SIDE VIEW

ACME-ECO-485 In Process

SIZE	CAGE CODE	DWG NO	REV
C	0SG68	S1101EFA0313	C
SCALE: NONE	INITIAL REL DATE 7/16/13	SHEET 2 OF 2	

BLUEPRINT
VERSION 3.0

REV C
SH 2
DWG NO S1101EFA0313